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Problem Statement: Smart Gate Access Control System

Current State of Operations

The University of Eswatini (UNESWA), as the premier institution of higher learning in the Kingdom, manages a high volume of daily traffic across its Kwaluseni and satellite campuses. Currently, campus gate management relies exclusively on manual oversight by security personnel. Guards are tasked with visually verifying identification and manually permitting entry or exit. While this human-centric approach provides a physical presence, it lacks the technical infrastructure required for modern institutional security.

Operational Inefficiencies and Security Risks

The absence of a digital tracking mechanism has led to several critical vulnerabilities:

- **Lack of Verifiable Audit Trails:** There is currently no recorded history of who enters or exits the campus, at what time, or through which gate. In the event of a security breach or criminal investigation, the university has no data to reconstruct timelines or identify individuals present on site.
- **Identity Verification Failures:** Manual verification is prone to human error. Without a scanning system, it is easy for unauthorized individuals to use expired, forged, or borrowed identification cards to gain entry.
- **Property and Asset Vulnerability:** The university faces significant risks regarding the theft of institutional and personal property. Without a system to log the movement of assets (such as laptops or university equipment) in association with a user's digital profile, tracking stolen goods becomes nearly impossible once they leave the perimeter.
- **Congestion and Response Latency:** During peak hours, manual checks create bottlenecks at entry points. Furthermore, security personnel lack a ”real-time view” of campus occupancy, which is vital for emergency response and crowd management.

The Data Gap

From a management perspective, the lack of structured data prevents informed decision-making. The university administration cannot currently analyze traffic patterns, determine peak usage times for specific gates, or allocate security resources based on empirical evidence. This ”data blindness” results in inefficient staffing and leaves the campus exposed to unpredictable security threats.

The Proposed Solution

To address these challenges, there is an urgent need for the **Smart Gate Access Control System**. By transitioning from a manual-only process to a hybrid, technology-driven model, UN-ESWA will digitize its security operations Implementing Java-based barcode/QR scanning integrated with a MySQL relational database will ensure that every movement is authenticated, timestamped, and logged. This transition is not merely an upgrade in convenience but a necessary modernization to safeguard the university’s people, property, and reputation